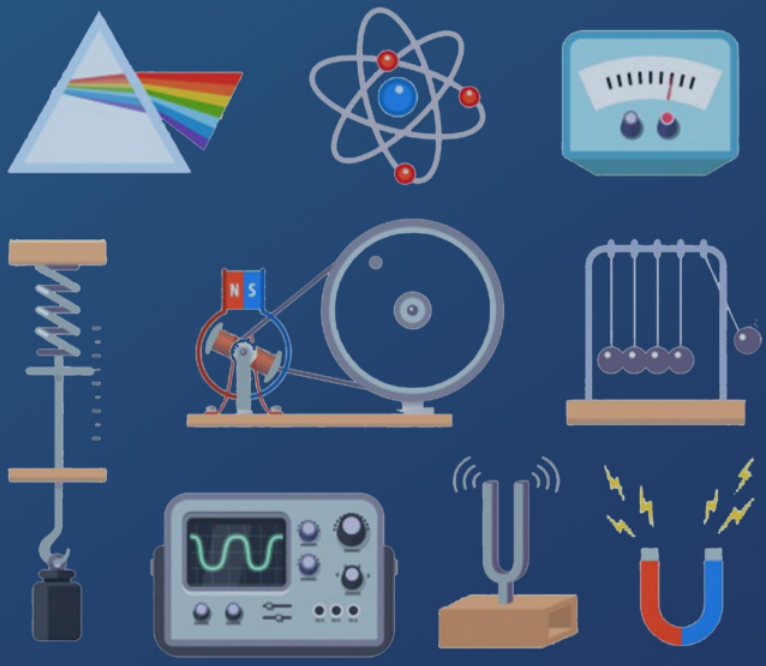
PHYSICS

X-ray



SHEET

eSaral.com

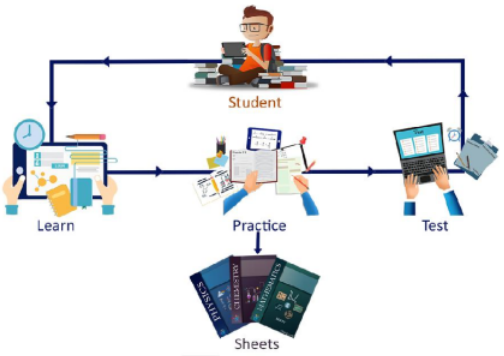
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Coolidge Tube	1,2			5
2	Continuous X-Rays	3,4			1
3	Characteristic X-Rays	5-7	1,2	1	6
4	Moseley's Law			2-4	2,7,8
5	Bragg's Law				3,4

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Coolidge Tube

Q1. Which of the following are the characteristics required for the target to produce X–rays

Option :	(A)	(B)	(C)	(D)
Atomic number	Low	High	Low	High
Melting point	High	High	Low	Low

Q2. In a Coolidge tube experiment, the minimum wavelength of the continuous X–ray spectrum is equal to 66.3 pm, then

(A) electrons accelerate through a potential difference of 12.75 kV in the Coolidge tube

(B) electrons accelerate through a potential difference of 18.75 kV in the Coolidge tube

(C) de–Broglie wavelength of the electrons reaching the anticathode is of the order of 10 μm

(D) de–Brgolie wavelength of the electrons reaching the anticathode is 0.01 Å

Continuous x-rays

Q3. On operating an X–ray tube at 1 kV. X–rays of minimum wavelength 6.22 Å are produced. If the tube is operated at 10 kV, then the minimum wavelength of X–rays produced will be

(A) 0.622 Å (B) 6.22 Å

(C) 3.11 Å (D) 62.2 Å

Q4. In an X-ray tube, electrons striking a target are brought to rest, causing X-rays to be emitted. In a particular X-ray tube, the maximum frequency of the emitted continuum X-ray spectrum is f_o. If the voltage across the tube is doubled, the maximum frequency is

(A) $f_o / 2$ (B) $f_o / \sqrt{2}$

(C) f_o (D) $2f_o$

Characteristic x-rays

Q5. The K_α X–ray emission line of tungsten occurs at $\lambda = 0.021$ nm. The energy difference between K and L levels in this atoms is about :

(A) 0.51 MeV (B) 1.2 MeV

(C) 59 keV (D) 13.6 eV

Q6. The production of characteristic X–rays is due to–

(A) transfer of momentum in collision of electrons with the target atom

(B) transfer of energy in collision of electrons with the target atom

(C) the transition of electrons in heavy target atoms from high to low energy level

(D) none of these

Q7. Characteristic X–rays are not obtained in the spectrum of H–atom because–

(A) hydrogen is a gas

(B) hydrogen is very light

(C) energy difference in energy levels of hydrogen is much less

(D) energy difference in energy levels of hydrogen is much high

Exercise - 1A — Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Which of the following is/are not true for x-rays ?

(A) Wavelength of continuous x-rays does not depend on potential difference

(B) Wavelength of discrete x-rays does not depends on potential difference

(C) Discrete x-rays have energy of the order of MeV

(D) Continuous x-rays have energy of the order of KeV

ASSERTION AND REASON

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q2. **Statement–1 :** The frequency of K_α X–radiation is greater than K_β for a given target material.

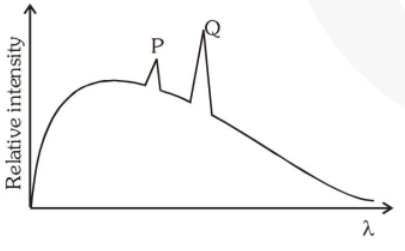
Statement–2 : K_α radiation is produced when an electron from n = 2 jumps into the vacancy in n = 1 orbit; whereas in K_β radiation the transition takes place from n = 3 to n = 1.

Exercise - 2 — Brain Booster

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. In a characteristic X–ray spectra of some atom superimposed on continuous X–ray spectra



(A) P represents K_α line

(B) Q represents K_β line

(C) Q and P represents K_α and K_β lines respectively

(D) Relative positions of K_α and K_β depends on the particular atom

Q2. The wavelength of K_α line for an element of atomic number 43 is λ. Then the wavelength of K_α line for an element of atomic number 29 is :–

(A) (43/29) λ (B) (42/28) λ

(C) (9/4) λ (D) (4/9) λ

Q3. The wavelengths of K_α X–rays of two metals 'A' and 'B' are $\frac{4}{1875R}$ and $\frac{1}{675R}$ respectively, where 'R' is Rydberg constant. The number of elements lying between 'A' and 'B' according to their atomic numbers is :

(A) 3 (B) 6 (C) 5 (D) 4

Q4. The constant 'a' in Mosely's law $\sqrt{\nu} = a(Z-b)$ for K_α X-ray is related to Rydberg constant R as ('c' is the speed of light)

(A) $R = \frac{4a^2}{3c}$ (B) $R = \frac{3c}{4a}$

(C) $R = \frac{3c^2}{4}$ (D) $R = \frac{2c}{a}$

ONE OR MORE THAN ONE CORRECT TYPE

Q1. The potential difference applied to an X–ray tube is increase. As a result, in the emitted radiation

(A) the intensity increases

(B) the minimum wavelength increases

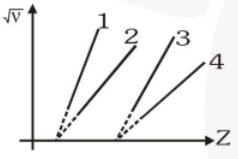
(C) the intensity decreases

(D) the minimum wavelength decreases

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

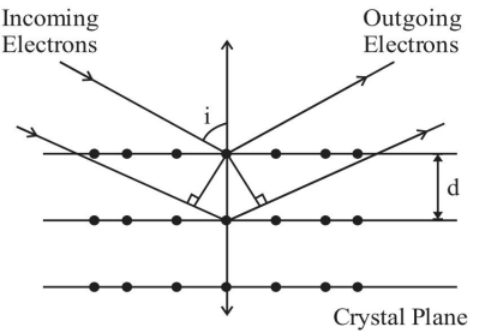
Q2. $\sqrt{\nu}$ versus Z graph for characteristic X–rays is as shown in figure. Match the following :



	Column I	Column II
(A)	Line – 1	(p) L_α
(B)	Line – 2	(q) L_β
(C)	Line – 3	(r) K_α
(D)	Line – 4	(s) K_β

Paragraph for Question No. 3 to 4

Wave property of electrons implies that they will show diffraction effects. Davisson and Germer demonstrated this by diffracting electrons from crystals. The law governing the diffraction from a crystal is obtained by requiring that electron waves reflected from the planes of atoms in a crystal interfere constructively (see figure).



Q3. Electrons accelerated by potential V are diffracted from a crystal.

If d = 1 Å and i = 30°, V should be about

($h = 6.6 \times 10^{-34}$ J-s, $m_e = 9.1 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C)

(A) 2000 V (B) 50 V

(C) 500 V (D) 1000 V

Q4. If a strong diffraction peak is observed when electrons are incident at an angle i from the normal to the crystal planes with distance d between them (see figure), de-Broglie wavelength λ_{dB} of electrons can be calculated by the relationship

(n is an integer)

(A) $d \sin i = n\lambda_{dB}$ (B) $2d \cos i = n\lambda_{dB}$

(C) $2d \sin i = n\lambda_{dB}$ (D) $d \cos i = n\lambda_{dB}$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q5. The K_α X–ray of molybdenum has wavelength 71pm. If the energy of a molybdenum atom with a K electron knocked out is 23·32 keV, What will be the energy of this atom when an L–electron is knocked out?

Q6. If three times of λ_{min} of continuous X–ray spectrum of target metal at 40 kV is same as the wavelength of K_α line of this metal at 30 kV, then determine the atomic number of the target metal .

SUBJECTIVE TYPE

Q7. An X–ray tube, operated at a potential difference of 40 kV, produces heat at the rate of 720 W. Assuming 0.5% of the energy of incident electrons is converted into X–rays, calculate

(i) the number of electron per second striking the target.

(ii) the velocity of the incident electrons

Q8. When X-rays of wavelength 0.5Å pass through 10 mm thick 'Al' sheet then thier intensity is reduced to one sixth. Find the absorption co-efficient for Al.

Answer Key

Ex-1						
1	2	3	4	5	6	7
B	B	A	D	C	C	C

Ex-1A

1. A,C,D **2.** D

Ex-2			
1	2	3	4
C	C	D	A

Ex-2A

1. A,D **2.** (A) s (B) r (C) q (D) p **3.** B **4.** B

5. 5.86 keV **6.** 37

7. (i) 1.1×10^{17} (ii) 1.2×10^{8} m/s **8.** 0.18 mm^{-1}